

Strengthening India's Critical Minerals Value Chain: Reducing Import Dependence and Enhancing Resilience

Focusing on the strategic and economic importance of critical minerals, this brief examines India's import vulnerabilities and midstream processing gaps, proposing integrated measures to develop domestic capacity, foster innovation, and ensure long-term supply security.

EXECUTIVE SUMMARY

India's growing dependence on critical minerals—such as lithium, cobalt, nickel, and rare earth elements—poses significant economic and strategic risks amid accelerating energy transition, digitalisation, and industrial expansion. Despite identifying 30 critical minerals, the country remains heavily import-dependent, with near-total reliance on several key inputs. This vulnerability is amplified by the concentration of global supply chains, particularly in processing and refining.

The analysis highlights a structural imbalance: rising demand driven by EVs, renewable energy, and PLI-supported sectors contrasts with weak domestic capabilities in processing, technology commercialisation, and value chain integration. The absence of a robust midstream segment leads to continued reliance on high-value imports.

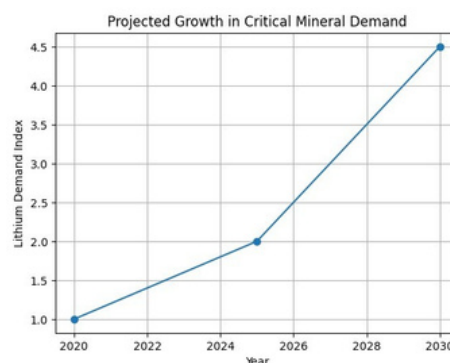
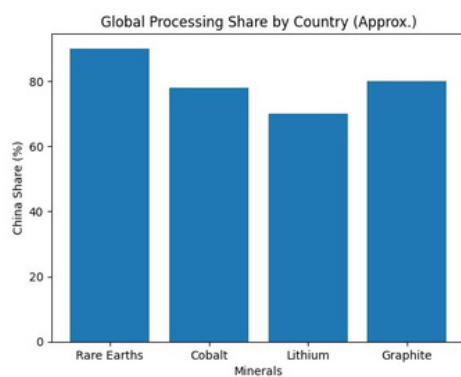
To address these gaps, the brief recommends developing processing parks, strengthening innovation through a dedicated fund, expanding fiscal incentives for refining, scaling up recycling, building strategic reserves, and enhancing global partnerships.

These measures aim to build a self-reliant critical minerals ecosystem.

CONTEXT AND SIGNIFICANCE

India's shift toward a high-growth, technology-driven, and low-carbon economy has made access to critical minerals—lithium, cobalt, nickel, rare earth elements (REEs), and graphite—strategically essential. These materials underpin electric vehicles, renewable energy, electronics, and defence, positioning them at the centre of India's energy transition, digitalisation, and net-zero 2070 goals. At the same time, global demand is rising sharply, with lithium alone projected to grow 4–5 times by 2030. Supply chains, however, remain highly concentrated—particularly in processing, where China dominates—creating significant geopolitical and supply risks for import-dependent economies like India.

Domestically, these risks are amplified by limited production and weak midstream capacity. Despite identifying 30 critical minerals, India remains heavily reliant on imports for key resources and lacks sufficient processing and refining infrastructure, restricting value addition. While recent initiatives signal policy intent, the focus has largely remained on downstream manufacturing. The core challenge lies in building an integrated value chain across exploration, processing, and recycling. Without this, India faces continued exposure to supply disruptions, price volatility, and loss of economic value. Developing a resilient, self-reliant critical minerals ecosystem is therefore essential to sustaining growth, enabling clean energy transitions, and strengthening strategic autonomy.



POLICY LANDSCAPE

India's critical minerals strategy has evolved with the **National Critical Mineral Mission (NCMM)**, backed by ₹34,300 crore, shifting from fragmented efforts to a coordinated, whole-of-government approach across the value chain—exploration, mining, processing, recycling, and recovery. Framing minerals as strategic assets for energy security, industrial growth, and defence, the policy introduces measures such as strategic stockpiling, fast-track approvals, and a Centre of Excellence to strengthen long-term resource security.

Alongside this, market-oriented reforms—through amendments to the MMDR and Offshore Areas Mineral Acts—have improved transparency via auction-based allocations, increased private participation, and enabled a more competitive, technology-driven sector. Recognising limited domestic reserves, India is also pursuing global partnerships through KABIL while investing in processing infrastructure, recycling, and circular economy models to boost domestic value addition and reduce import dependence. Together, these efforts mark a shift toward a more resilient and integrated critical minerals ecosystem.

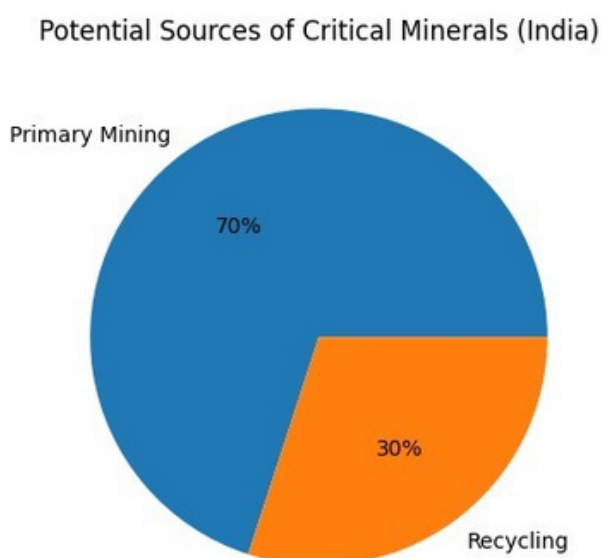
KEY CHALLENGES

India's primary structural constraint is the midstream processing gap—despite some extraction capacity, it lacks advanced refining for high-purity inputs, while China controls over 90% of rare earth processing, ~80% of cobalt refining, and ~70% of lithium chemicals. This gap extends to quality: India's graphite purity (92–99%) falls short of the 99.95% required for EV batteries, and rare earth separation into high-purity elements is absent, creating a paradox of exporting raw materials and importing high-value finished products. Additional challenges include: weak commercialisation of technology (research stuck at pilot/lab stages); skill and capability mismatch (workforce geared to traditional mining, lacking hydrometallurgy & chemical refining); investment constraints (high risk, uncertain returns, policy ambiguity, weak auction participation); regulatory fragmentation (multiple approvals, overlapping mandates, delays, lack of demand assurance); and environmental & social barriers (resistance over displacement, ecological damage, higher costs).

POLICY RECOMMENDATIONS

To address India's critical minerals gaps—particularly in midstream processing and import dependence—the following measures are recommended. First, establish 3–4 integrated Critical Minerals Processing Parks with plug-and-play infrastructure for lithium, cobalt, rare earth elements, and graphite, including shared hydro-metallurgy, testing, and waste treatment facilities. This will reduce capital costs, promote cluster-based ecosystems, and directly fill the midstream gap. Second, create a ₹5,000–₹10,000 crore Lab-to-Market Innovation Fund to support pilot plants, demonstration projects, and industry-academia collaboration, bridging technology readiness levels and enabling commercialisation. Third, provide targeted fiscal incentives—capital subsidies, tax benefits, concessional financing—and extend the PLI scheme to midstream processing, aligning mineral policy with industrial objectives. Fourth, launch a National Urban

Mining and Recycling Mission to recover minerals from e-waste, spent batteries, and industrial waste, with mandatory recycling targets, battery take-back systems, and formal recycling hubs, leveraging secondary sources as a future supply. Fifth, build strategic reserves of high-risk minerals like lithium, cobalt, and rare earth elements, modelled on Strategic Petroleum Reserves, to buffer against supply disruptions and price volatility. Sixth, strengthen global partnerships through KABIL, the Mineral Security Partnership, and bilateral agreements with resource-rich countries, focusing on joint ventures in processing and refining for greater value addition. Finally, establish a Centralised Critical Minerals Authority to unify oversight across exploration, processing, recycling, and international engagements, reducing fragmentation, streamlining approvals, and enhancing investor confidence.



CONCLUSION

India's critical minerals sector is constrained by high import dependence and limited domestic processing capacity, creating economic and strategic vulnerabilities. With rising demand from clean energy and advanced manufacturing, addressing these gaps is increasingly urgent.

Three key actions are essential. First, develop **critical minerals processing** parks to strengthen domestic refining capacity. Second, create an **innovation fund** to support the transition of technologies from lab to industry. Third, enhance **global partnerships** and strategic reserves to ensure stable and secure supply.

These steps will help build a more resilient and self-reliant critical minerals ecosystem for India's long-term growth.

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